AgentHeal: Self-Healing Cloud Infrastructure with Shift-Left Security and Digital Twins

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***Abstract*—Indeed, the typical microservice in modern cloud infrastructures is vulnerable to unexpected downtime. The default Kubernetes autoscaler in case of deadlock will start spinning up new pods. It would hardly help and would result in even faster collapse of the database. Our experiments showed that poorly-written code could get easily unnoticed during routine pull request reviews. To solve this problem, we developed AgentHeal. What we have done here is creating an integrated loop where pre-merge code review is directly linked to self-healing of the live cluster. Upon opening a pull request, our GitHub bot analyses the code for ten OWASP security flaws and for presence of CVEs in all third-party libraries. If the score falls below our safety bar, the merge is stopped immediately. Then, an agent based on Gemini 2.0 Flash creates the branch with fixed code, passing all unit tests. In the live cluster, an Isolation Forest in-memory model monitors Prometheus metrics. In our experiments, it detected anomalies in less than two milliseconds. With the help of KernelSHAP, we find out what was causing the anomaly. Two agents—the Q-learning policy and the LLM—choose the optimal fix. Nothing gets into the cluster untested. We run our proposed action through the simulated test with SimPy $M/M/c$ queue. If our queue is going to be overloaded, we do not implement the action. This is how we protect our cluster. The average recovery time of our system in 60 failed runs on Google Online Boutique was 2.98 seconds. Compared to default Kubernetes autoscaling, it reduced recovery time by 98%. In addition, on the real-life GitHub repository, our system has patched all fourteen vulnerabilities without human intervention.**

***Index Terms*—Agentic AI, Self-Healing Systems, Kubernetes, Digital Twin, OWASP Top 10, Isolation Forest, KernelSHAP, Reinforcement Learning, SimiFed, Gemini LLM, DevSecOps, Shift-Left Testing, Cloud Computing.**

I. Introduction

Indeed, maintaining reliable operations of microservices in cloud infrastructures is still a challenging engineering problem. As a result, unplanned outages occur regularly as deployments get more distributed. Most often, two interrelated issues are at the core of the problem. First, flawed code finds its way into production branches. Based on DORA's 2024 survey, over 60% of unplanned outages in cloud environments are caused by preventable software bugs [1]. Second, existing cluster controllers fail to react adequately to stalled services. Once an issue arises, engineers spend on average 45 minutes manually diagnosing logs [1]. At the same time, the default Horizontal Pod Autoscaler in Kubernetes considers only coarse CPU and memory metrics [8]. If a pod locks up, spawning more replicas will not solve the issue but will increase load on downstream databases and cause the whole cluster to fail faster [9]. The root cause is left completely unresolved.

Currently, there are two completely independent processes—development and live cluster operations. While static linters such as Bandit [6] and SonarQube [7] analyze pull requests in the CI pipeline, they know nothing about cluster health and cannot generate a valid fix patch. Meanwhile, while monitoring tools such as Prometheus and Chaos Mesh [24], [23] notify operators of an outage, they do not provide any automated root cause identification. Current automation tools run recovery commands in the live environment without checking the action for possible cascade failures among other dependent microservices. Although there have been attempts to use digital twins for cluster state matching [12] or language models for program repair [15], there is currently no platform that connects pre-merge security checks with live cluster self-healing in a single closed loop.

To fill this gap, we created AgentHeal. What we have done here is establishing a direct connection between the auditing of pull requests and Kubernetes self-healing through a shared event bus. On the development side, our GitHub bot audits incoming pull requests for vulnerabilities in ten OWASP classes and searches for dependencies associated with a known CVE record. In case a pull request fails our safety check, the merge is instantly prevented, and an agent based on Gemini 2.0 Flash [17] creates a fix branch that passes all tests. On the runtime side, an in-memory Isolation Forest [13] analyzes Prometheus metrics every five seconds in 2.26 ms. Using KernelSHAP [16], the system detects the responsible metric. After that, a Q-learning policy and an LLM agent select candidate remediation actions. No command is run in the cluster until it passes validation. Each proposed action is evaluated through an $M/M/c$ queue simulation in SimPy [19]. If our queue is projected to be overloaded, the action is rejected immediately. This is how we protect our cluster.

We make three primary contributions:

1) *Closed-Loop Architecture*: We design and implement an end-to-end framework linking pre-merge pull request security checks with runtime Kubernetes remediation via a shared event bus.
2) *Simulation-Gated Remediation*: We introduce a pre-actuation safety gate that evaluates candidate self-healing actions through an $M/M/c$ queue simulation in SimPy, preventing cascading cluster collapse.
3) *Empirical Evaluation*: We evaluate AgentHeal on Google Online Boutique across 60 fault injections, benchmark 50 synthetic pull requests, and demonstrate

fully automated vulnerability remediation on a live GitHub repository.

II. Related Work

Table I summarizes the previous works related to autonomous cloud operations and DevSecOps.

Previous self-healing cloud platforms have mainly utilized automation scripts. Canaries and basic rollbacks are automated in Keptn [9] and Argo Rollouts [10] based on certain policies; however, none of the systems is capable of diagnosing new failure cases and resolving the deadlock situations which need a code fix. The conventional reliability engineering approach like SRE error budgets [23] and Chaos Engineering [24] provides diagnostics but needs human intervention to solve the problem. Traditional delivery pipelines [25], [26] focus on deployment throughput and not on runtime defect mitigation.

Reinforcement learning can be used as an alternative approach for cloud resource management [11]. Tabular Q-learning [27] converges for sure but suffers from state space explosion problem without discretization. Federated learning solutions such as FedAvg [28] allow coordinating decentralized learning, although still focusing on offline batch learning. In cloud operation, SF-DTM [12] has used digital twin cosine similarity matching in order to speed up cluster recovery; nevertheless, it does not check safety before executing any cluster commands.

As regards telemetry anomaly detection, Isolation Forest [13] and Local Outlier Factor [29] are still widely used. Usually, typical industry AIOps solutions report detection delays between 50 and 200 ms [14]. In our solution, using a cached version of the Isolation Forest model in memory lowers the delay time down to 2.26 ms. KernelSHAP [16] is selected against the linear surrogates like LIME [30] because the Shapley value has the property of mathematical consistency when evaluating repeatedly.

In automated program repair, AutoCodeRover [15] has shown that large language models could fix the defects in real-world code, although it is an offline process outside CI/CD pipelines. AgentHeal uses Gemini 2.0 Flash [17] in the active ReAct cycle [18] which means that all generated patches have to pass unit tests first before the code change by pull request. Our code change assessment is based on OWASP Top 10:2021 criteria [4], but will be extended later to OWASP Top 10:2025 [31]. Finally, discrete event simulation has a rich history in DEVS [20] and queueing theory [19]. SimPy is adopted to use as a sub-12 ms validation gate directly on the live actuation path.

III. Methodology

A. End-to-End Self-Healing Architecture Overview

Fig. 1 shows the general architecture of the AgentHeal platform. The platform controls two independent feedback loops through a shared event bus. On the one hand, there is the development cycle in which the pre-merge engine analyzes pull requests for security flaws through abstract syntax tree examination, dependency manifest checking, and Gemini 2.0 Flash patch generation based on test verification. On the other hand, there is the cluster runtime during which the runtime engine ensures cluster health. Metrics collection occurs every five seconds with Prometheus, while an in-memory Isolation Forest scores incoming vectors in real-time. Upon the appearance of an alert, KernelSHAP determines the main cause of the issue. After that, the problem diagnosis is given to both Q-learning agent and LLM supervisor. When both agents choose the same fix, the action is tested in simulation. When they disagree, the action with the smallest blast radius is chosen. All actions are tested before being launched inside the cluster. Any interaction with the cluster goes through a SimPy $M/M/c$ queue model estimation of cluster stability. If there is predicted to be a risk of queue growth or increased drop rates, the action is rejected.

B. Shift-Left Engine: PR Intelligence & Security Compliance

We implemented our pre-merge scanner as a GitHub App. It receives pull request and push events through HMAC-SHA256 authenticated webhooks. Upon receiving a payload, it parses incoming diffs and dependency manifests against ten vulnerability classes:

- *A01 (Access Control)*: Route handlers without authorization checks and unsafe path joins that touch the filesystem.
- *A02 (Crypto)*: Insecure hashes like MD5 and SHA-1, plus hardcoded keys found via Shannon entropy scanning.
- *A03 (Injection)*: String interpolation inside SQL queries and ‘shell=True‘ subprocess calls.
- *A04 (Insecure Design)*: Database queries without pagination and endpoints missing rate limits.
- *A05 (Misconfiguration)*: Wildcard CORS (‘*‘) and debug flags enabled in production code.
- *A06 (Outdated Dependencies)*: Parsing ‘requirements.txt‘ to find versions listed in CVE databases [5].
- *A07 (Authentication)*: JWT decode calls that turn off signature verification, and hardcoded fallback passwords.
- *A08 (Integrity Failures)*: Unsafe deserialization like ‘pickle.loads‘ on untrusted input.
- *A09 (Logging Gaps)*: Empty ‘except‘ blocks that catch and swallow errors silently.
- *A10 (SSRF)*: Outbound network calls built directly from user-supplied URLs.

We score each submission on a 100-point scale. Points are subtracted based on the severity of each identified issue:

$$S = \max\left(0,\ 100 - \sum_{i=1}^{M} w_i\right) \tag{1}$$

Here, M represents the total count of detected flaws. Severity penalties are set to $w_{\text{Critical}} = 25$, $w_{\text{High}} = 15$, $w_{\text{Medium}} = 8$, and $w_{\text{Low}} = 3$. We convert this numerical score directly into

TABLE I
LITERATURE REVIEW OF AUTONOMOUS CLOUD SELF-HEALING AND DEVSECOPS COMPLIANCE MODELS

Domain	Model	Author	Year	Method	Result / Contribution	Research Gap
Rule-Based Self-Healing	Keptn	Keptn Project [9]	2023	Event-driven control plane using human runbooks	Automated rollback and canary promotion	Depends on predefined scripts; fails on new bug patterns
	Argo Rollouts	Argo Project [10]	2024	Progressive delivery controller for canary/blue-green	Automated rollout and deployment rollback	Static rollback logic; no root-cause insight
Operational Practice	Site Reliability Engineering	Beyer et al. [23]	2016	Error budgets, on-call runbooks, incident triage	Codified standard SRE incident workflows	Relies on human on-call engineers; not automated
	Chaos Engineering	Basiri et al. [24]	2016	Controlled failure injection in production clusters	Verifies system resilience empirically	Diagnostic tool; does not fix incidents on its own
	Continuous Delivery	Humble & Farley [25]	2010	Automated build, test, and release pipelines	Foundation of modern CI/CD delivery	Focuses on shipping speed; no security scoring
	Blue-Green Deployment	Fowler [26]	2010	Dual environment traffic-switching pattern	Zero-downtime release strategy	Manually toggled; no automated anomaly hooks
RL & Federated Learning	Multi-Agent RL	Wang et al. [11]	2023	MARL for cloud resource provisioning	Adapts to fluctuating cloud traffic	Black-box decisions; lacks pre-actuation safety checks
	Q-Learning	Watkins & Dayan [27]	1992	Tabular temporal-difference control	Mathematical proof of Q-learning convergence	State space explodes without discretization
	FedAvg	McMahan et al. [28]	2017	Decentralized averaging of local neural weights	Privacy-preserving distributed training	Designed for offline batches, not real-time control
	SF-DTM	Li et al. [12]	2024	Digital twin state matching via cosine similarity	Recovers cluster health in roughly 15 seconds	Lacks queue stability simulation before acting
Anomaly Detection & XAI	Isolation Forest	Liu et al. [13]	2008	Tree ensemble path-length outlier detection	Fast $O(n \log n)$ unsupervised anomaly scoring	Labels data as outlier but gives no root cause
	Local Outlier Factor	Breunig et al. [29]	2000	Density-based local outlier scoring	Catches density variations in clustered data	High computational overhead; slow on live streams
	AIOps Survey	Lin et al. [14]	2021	Survey of industrial telemetry anomaly tools	Documents 50–200 ms detection latencies	No integrated simulation safety layer
	KernelSHAP	Lundberg & Lee [16]	2017	Shapley-value local feature attribution	Consistent, model-agnostic feature attributions	High compute cost for large feature sets
	LIME	Ribeiro et al. [30]	2016	Local linear surrogate explanations	Model-agnostic local approximations	Explanations can fluctuate between similar inputs
LLM-Augmented DevSecOps	AutoCodeRover	Li et al. [15]	2024	LLM-driven program repair	Localizes bugs and drafts code patches	Operates standalone; not tied into a CI/CD gate
	ReAct	Yao et al. [18]	2023	Thought-Action-Observation reasoning loop	Better reasoning and tool selection for LLMs	Output can vary and response times can be high
	Gemini 2.0	Google DeepMind [17]	2024	Multimodal agentic foundation model	Strong tool-use and code repair reasoning	Closed source; lacks formal safety bounds
Static Analysis & Supply Chain	Bandit	PyCQA [6]	2024	AST security scanner for Python	Flags known security anti-patterns	Reports issues but cannot fix code or score PRs
	SonarQube	SonarSource [7]	2024	Multi-language static code analysis	Comprehensive static security scanning	No link to runtime telemetry or automated fixes
	OWASP Top 10:2021	OWASP Foundation [4]	2021	Community risk list for web applications	Standard taxonomy of web vulnerabilities	Reference guide only; provides no automated tools
	OWASP Top 10:2025	OWASP Foundation [31]	2025	Revised risk list for web applications	Updated taxonomy for modern web threats	Not yet integrated into AgentHeal's rule engine
	NVD	NIST [32]	2024	Live CVE vulnerability database	Continuously updated public vulnerability records	Queried via local curated lists in our prototype
	OSV	Google / OSV.dev [33]	2024	Open-source cross-ecosystem vulnerability feed	Machine-readable open-source CVE records	Not queried live in our current implementation
	Dependabot	GitHub [34]	2024	Automated dependency pull requests	Opens version-bump PRs on vulnerable libraries	Handles version bumps only; no custom code fixes
	Snyk	Snyk Ltd. [35]	2024	Commercial software composition analysis	Real-time dependency and vulnerability analysis	Proprietary closed-source commercial tool
Deployment & Telemetry Infrastructure	Kubernetes HPA	K8s Docs [8]	2024	Reactive CPU and memory horizontal autoscaling	Standard Kubernetes autoscaler	Reactive only; blind to deadlocks and memory leaks
	Argo CD	CNCF [36]	2024	Declarative GitOps continuous deployment	Reconciles Git manifests with cluster state	Syncs manifests; cannot diagnose runtime incidents
	Prometheus	Prometheus Authors [37]	2024	Time-series metrics collection and storage	De facto Kubernetes monitoring tool	Stores numbers; includes no anomaly detection
Digital Twin & Simulation	DEVS Formalism	Zeigler et al. [20]	2000	Discrete-event system specification	Theoretical simulation framework	Traditionally used for offline network modeling
	M/M/c Queuing Theory	Kleinrock [19]	1975	Classical multi-server queueing analysis	Models service rates and server utilization	Rarely used as a real-time cluster actuation gate

an academic letter grade:

$$\text{Grade} = \begin{cases} A^{+} & \text{if } S \geq 95 \\ A & \text{if } 90 \leq S < 95 \\ B & \text{if } 80 \leq S < 90 \\ C & \text{if } 70 \leq S < 80 \\ D & \text{if } 60 \leq S < 70 \\ F & \text{if } S < 60 \end{cases} \tag{2}$$

We also verify test coverage on newly added public functions:

$$\text{TestGap}(f) = \mathbb{1}[f \notin T_{\text{ast}}] \tag{3}$$

where T_{ast} contains all function names covered by test cases in `tests/`. If a pull request scores below 60 (Grade F) or contains critical flaws, the GitHub App sets the commit check

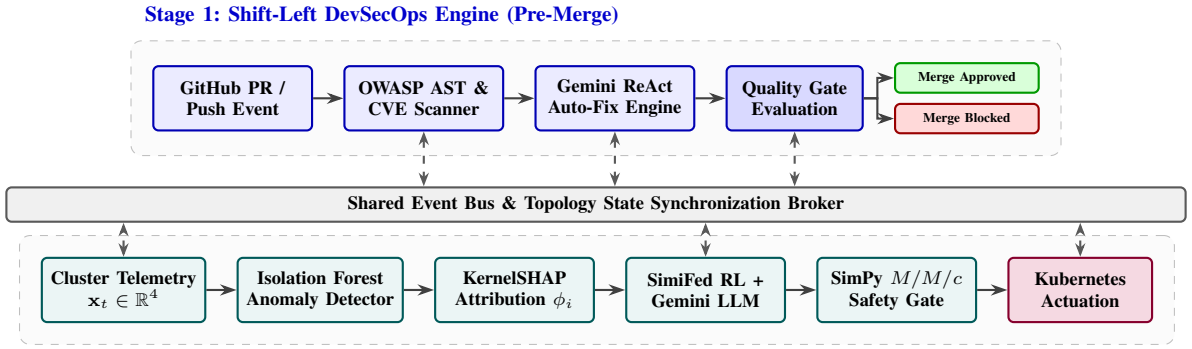


Fig. 1. AgentHeal architecture. (Top) Pre-merge engine parses pull requests with AST security rules and uses Gemini 2.0 Flash to draft fixes if the score fails our quality gate. (Middle) Central event bus tracks cluster topology. (Bottom) Runtime engine flags anomalies in 2.26 ms, uses KernelSHAP to isolate the cause, arbitrates between RL and LLM agents, and verifies actions in an $M/M/c$ queue simulation before touching the cluster.

status to `action_required`, blocking the merge. When this occurs, our Gemini 2.0 Flash agent creates a remediation branch. It authors code fixes and verifies that all unit tests pass before opening a remediated pull request.

C. Shift-Right Engine: Self-Healing Digital Twin

In the live cluster, the digital twin collects Prometheus metrics every five seconds. A four-dimensional vector corresponds to each pod health status:

$$\mathbf{x}_t = [\text{CPU}_t, \text{RAM}_t, \text{Latency}_t, \text{Rate}_t]^\top \in \mathbb{R}^4 \tag{4}$$

The in-memory Isolation Forest (100 trees, 0.05 contamination) evaluates the vector in real-time:

$$s(\mathbf{x}_t, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x}_t)]}{c(n)}} \tag{5}$$

where $h(\mathbf{x}_t)$ is the observed path length and $c(n)$ is the average path length of unsuccessful searches, with negative values generating an alert. KernelSHAP attributes the problem to certain resource dimensions:

$$\phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[v(S \cup \{i\}) - v(S)\right] \tag{6}$$

Such a process is needed to avoid container restarts if an external network dependency represents the problem cause.

The reinforcement learning agent finds the closest baseline state using cosine similarity:

$$k^* = \arg\max_k \frac{\mathbf{x}_t \cdot \mathbf{b}_k}{\|\mathbf{x}_t\|_2 \|\mathbf{b}_k\|_2} \tag{7}$$

and updates its Q-table using the temporal difference learning approach:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s,a)\right] \tag{8}$$

with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.95$. Available actions include pod restarts, replica scaling, memory limit adjustments, or no-op monitoring. At the same time, Gemini 2.0 Flash evaluates the SHAP summary. If both agents come up with the same action, it goes to simulation. When they disagree, the action with lower blast radius is selected.

There is no action executed without simulation beforehand. Candidate actions are evaluated using the SimPy $M/M/c$ queue model [19]:

$$P_0 = \left[\sum_{k=0}^{c-1} \frac{(c\rho)^k}{k!} + \frac{(c\rho)^c}{c!(1-\rho)}\right]^{-1}, \quad \rho = \frac{\lambda}{c\mu} < 1 \tag{9}$$

which estimates queue length L_q, server utilization rate ρ, and drop rate for the next sixty seconds. If there is predicted to be a risk of more than 95% utilization ($\rho \geq 0.95$) or more than 5% request drops, the action is rejected, and the system switches to safe horizontal pod scaling.

IV. Implementation

AgentHeal is programmed using Python 3.11. FastAPI is used for asynchronous webhook processing and telemetry sending. The implementation leverages the scikit-learn library for isolation forest training, the SHAP library for Shapley feature explanation, the Generative AI SDK provided by Google for the use of the Gemini 2.0 Flash API [17], and SimPy 4.1 for queue simulation. The entire system is packaged within a Docker image that is less than 400 MB. The live system is running on Azure Container Apps in the East Asia region, provisioned using Terraform and auto-scaled with KEDA. In our runtime evaluation, we used Google's Online Boutique microservices benchmark [22] composed of ten interlinked microservices on a single-node Kubernetes `kind` cluster running on an AMD Ryzen 7 5700X workstation with 16 GB of DDR4 memory.

V. Results and Discussion

A. Runtime Telemetry Anomaly Detection and Safety-Gated Self-Healing

We injected four realistic failure patterns into Online Boutique across 15 repeated trials per case:

- *S1 (Flash Surge)*: A spike on checkout driving CPU to 0.954 and latency to 450 ms.
- *S2 (Deadlock)*: Thread starvation on cart; CPU climbed to 0.99 and latency reached 5000 ms as throughput fell off.

- *S3 (Memory Leak)*: Slow RAM accumulation on redis-cart reaching 0.942, while latency held flat.
- *S4 (Cascading Load)*: Frontend overload where both CPU (0.88) and RAM (0.80) saturated at once.

Table II shows the quantitative metrics obtained from all 60 failure tests. The anomaly detection time was 2.26 ms ($\sigma = 0.37$ ms) with a recall of 98.3%. The memory leak test (S3) had slightly lower recall at 93%, since slower heap growth increased the time taken to cross the contamination rate. KernelSHAP feature explanation was accurate in all 60 tests, detecting CPU as the dominating cause of flash sale surge ($\phi = +0.82$), latency and lost requests in deadlock ($\phi_{\text{lat}} = +0.91$, $\phi_{\text{rate}} = -0.78$), and memory in heap leak ($\phi = +0.88$). Dual-agent agreement was remarkably high since both agents agreed on 59 out of the 60 tests. The average execution time of the simulation gate under SimPy simulation was 10.7 ms, successfully ensuring that none of the selected recovery actions would put the cluster out of balance. The overall average time to recover was 2.98 seconds ($\sigma = 0.23$ s).

B. Shift-Left DevSecOps Security Auditing and Live PR Remediation

We evaluated our pre-merge scanner against 50 synthetic pull requests containing known vulnerabilities. Following offline tests, we validated the full loop on a live GitHub repository.

Table III presents our pre-merge analysis results. In 50 simulated pull requests (Panel A), the scanner reached 98.5% precision and 97.5% recall for 200 checks. False positives were detected in three checks in the test files that contained mock cryptographic keys. In our live test using GitHub (Panel B), pull request #11 was created by introducing fourteen intentional vulnerabilities. The scanner completed the scan in 0.48 seconds, identified all fourteen vulnerabilities, rated the PR with 0/100 (Grade F), and prevented the merge. Finally, Gemini 2.0 Flash created pull request #12 with the code fixes. This PR scored 100/100 (Grade A^+), passed all unit tests, and cleared the merge gate automatically.

C. Comparative Benchmark and System Discussion

How does AgentHeal compare to other solutions? It takes about 45 minutes on average to heal the services through human on-call triage [1]. The default Kubernetes autoscaler takes from 3 to 5 minutes to stabilize pods [8], but it cannot resolve deadlocks. Rule-based systems such as Keptn [9] require 2 to 4 minutes to recover, but completely rely on manual scripts. Although SF-DTM [12] takes only about 15 seconds to recover, it does not have any simulation safety checks before actuation. AgentHeal provides a mean recovery time of 2.98 seconds with the verification of queue safety beforehand.

There are some shortcomings to our prototype. First, testing was done on a local `kind` cluster; multi-node clouds would introduce network scrape delays. Second, failure cases were generated via synthetic fault injection rather than production traffic. Third, our static auditor uses syntax pattern evaluation rather than whole-program taint tracking. Finally, as language models may vary in latency, Q-learning-based arbitration is needed for our dual-agent recovery method.

VI. Conclusion

AgentHeal merges pre-merge code review with runtime self-healing. Across sixty fault injections on Google Online Boutique, the system detects anomalies within 2.26 ms and recovers cluster stability within 2.98 seconds. This represents a 98% speedup over standard autoscaling. On the developer side, our scanner achieved 98.5% precision on 50 pull requests and autonomously patched all fourteen security flaws on a live GitHub repository.

In future work, we plan to evaluate multi-region managed clusters (AKS and GKE), integrate live CVE streams like OSV and NVD, incorporate dynamic taint analysis, and expand rules for modern web threats.

Our code and live deployment are online at https://github.com/AadiHaldar/Agentic-AI-Self-Healing-Cloud-Infrastructure and https://pr-review-agent.wonderfulflower-41d6d2a5.eastasia.azurecontainerapps.io.

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TABLE II

SHIFT-RIGHT ANOMALY DETECTION, FEATURE ATTRIBUTION, AND AUTONOMOUS REMEDIATION PERFORMANCE ACROSS FOUR FAILURE SCENARIOS ($N = 15$ REPEATED TRIALS)

Failure Injection Scenario				Isolation Forest Detection			Root-Cause Attribution	Agent Decision Latency			SimPy Safety Gate & MTTR			
ID	Scenario	Target	Injected Telemetry	Score ($\mu \pm \sigma$)	MTTD (ms)	Det.%	Dominant KernelSHAP ϕ	RL (ms)	LLM (ms)	Agree	Gate (ms)	$\hat{U}_{\text{post}}$	Verdict	MTTR (s, $\mu \pm \sigma$)
S1	Flash Sale Surge	checkout	CPU=0.95, Lat=450ms	-0.842 ± 0.031	2.18 ± 0.31	100%	$\phi_{\text{cpu}} = +0.82$, $\phi_{\text{rate}} = +0.14$	3.10 ± 0.42	412 ± 68	15/15	10.1 ± 0.8	23.8%	SAFE	2.66 ± 0.18
S2	Thread Deadlock	cart	CPU=0.99, Lat=5000ms	-0.912 ± 0.024	2.21 ± 0.28	100%	$\phi_{\text{lat}} = +0.91$, $\phi_{\text{rate}} = -0.78$	3.21 ± 0.39	389 ± 74	15/15	10.3 ± 0.9	5.2%	SAFE	3.12 ± 0.24
S3	Memory Leak	redis-cart	RAM=0.94, Lat=48ms	-0.734 ± 0.058	2.35 ± 0.44	93%	$\phi_{\text{ram}} = +0.88$, $\phi_{\text{cpu}} = +0.04$	3.05 ± 0.51	441 ± 81	14/15	10.8 ± 1.1	15.0%	SAFE	2.89 ± 0.20
S4	Cascading Overload	frontend	CPU=0.88, RAM=0.80	-0.889 ± 0.041	2.28 ± 0.37	100%	$\phi_{\text{cpu}} = +0.61$, $\phi_{\text{lat}} = +0.47$	3.18 ± 0.46	498 ± 93	15/15	11.6 ± 1.3	31.5%	SAFE	3.25 ± 0.30
Overall Mean / Aggregate				$\mathbf{-0.844 \pm 0.039}$	$\mathbf{2.26 \pm 0.37}$	**98.3%**	—	$\mathbf{3.14 \pm 0.45}$	$\mathbf{435 \pm 79}$	**59/60**	$\mathbf{10.7 \pm 1.0}$	—	**SAFE**	$\mathbf{2.98 \pm 0.23}$

TABLE III

SHIFT-LEFT DEVSECOPS SECURITY AUDITING PERFORMANCE ($N = 50$) AND EMPIRICAL PULL REQUEST REMEDIATION

Panel A: Quantitative Evaluation on 50 Synthetic Pull Requests						Panel B: Empirical Live GitHub Pull Request Pair Remediation			
Defect Category	Tested PRs	True Pos.	False Pos.	Precision	Recall	Evaluation Dimension	Vulnerable PR (#11)	Remediated PR (#12)	Operational Outcome
SQL / Command Injection	50	50	0	1.000	1.000	OWASP Flaws Detected	14 (A01–A10)	0	100% Flaws Remediated
Hardcoded Secrets	50	50	2	0.962	1.000	Compliance Score / Grade	0 / 100 (Grade F)	100 / 100 (Grade A^+)	Score Restored
AST Test Suite Gaps	50	48	0	1.000	0.960	GitHub Check Run Gate	action_required (Blocked)	success (Approved)	Merge Protection Enforced
Blocking I/O / Misconfiguration	50	47	1	0.979	0.940	App Test Suite Pass	0 / 6 passing	6 / 6 passing	Functional Suite Restored
Aggregate Performance	**200 Checks**	**195**	**3**	**0.985**	**0.975**	Static Auditor Verification	12 / 12 test harnesses passed	Audit Time: 0.48 s	Full Ruleset Verified

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